# Database PROJECT

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ADVENTURE WORKING
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# Introduction / Background

Adventure Working is a dynamic and multifaceted company that operates across various core business functions, including production, human resources, purchasing, and sales. With a strong emphasis on operational efficiency and data-driven decision-making, the company maintains comprehensive records across all departments.

Its well-structured database captures detailed information about personnel, supply chains, product output, and customer transactions, reflecting an integrated approach to enterprise management. Adventure Working's commitment to maintaining accurate and organised data supports its strategic goals, enabling the company to respond effectively to internal needs and market demands.

# Normalisation Steps Taken

## 1NF (First Normal Form) – Atomicity

To meet 1NF, we ensured:

- All attributes hold atomic (indivisible) values.
- No repeating groups or arrays.

### Examples:

- Address is not split into multiple lines (e.g., Street, City, ZIP) in this version for simplicity, but can be if required.
- Phone and Email are stored as individual fields, not comma-separated lists.

## 2NF (Second Normal Form) – Eliminate Partial Dependencies

To meet 2NF, we ensured:

- The database is already in 1NF.
- Every non-key attribute fully functionally depends on the entire primary key, especially for tables with composite primary keys (like junction tables).

### Examples:

- In SalesOrder (SalesOrderID, ProductID), attributes like Quantity and UnitPrice depend on both SalesOrderID and ProductID, not just one.
- The Employee table has attributes like HireDate and Position that depend only on EmployeeID, not a partial key.

## 3NF (Third Normal Form) – Eliminate Transitive Dependencies

To meet 3NF, we ensured:

- The database is in 2NF.
- No non-prime attribute depends on another non-prime attribute (i.e., no transitive dependencies).

### Examples:

- In the Product table, CategoryID is a foreign key to the ProductCategory table. This removes transitive dependency on CategoryName and Description.
- The Department table stores the DepartmentName and links to a ManagerID, a foreign key to Employee, keeping each piece of information in its logical place.

### Justification for 3NF:

The database design satisfies Third Normal Form because:

- All attributes in every table directly depend on the primary key and nothing else.
- No duplicate data is stored across unrelated entities. For instance:
    - Supplier details are only stored in the Supplier table.
    - Product categories are isolated in a Category table rather than being repeated in the Product table.

# Logical Design (Data Dictionary & Object Descriptions)

## AdventureWorks Data Dictionary

| Table Name | Column Name | Data Type | PK/FK | Constraints | Description |
|---|---|---|---|---|---|
| Person.Person | BusinessEntityID | INT | PK | NOT NULL, Identity | Unique identifier for the person. |
| Person.Person | FirstName | NVARCHAR(50) | | NOT NULL | First name of the individual. |
| Person.Person | LastName | NVARCHAR(50) | | NOT NULL | Last name of the individual. |
| Person.Person | ModifiedDate | DATETIME | | NOT NULL | Last date the record was updated. |
| Sales.Customer | CustomerID | INT | PK | NOT NULL, Identity | Unique identifier for each customer. |
| Sales.Customer | PersonID | INT | FK | NULLABLE | Links to the Person table if individual customer. |
| Sales.Customer | StoreID | INT | FK | NULLABLE | Links to a store. |
| Sales.Customer | TerritoryID | INT | FK | | Links to the sales territory. |
| Sales.SalesOrderHeader | SalesOrderID | INT | PK | NOT NULL, Identity | Main identifier for a sales order. |
| Sales.SalesOrderHeader | CustomerID | INT | FK | | Identifies the customer placing the order. |
| Sales.SalesOrderHeader | OrderDate | DATETIME | | NOT NULL | The date the order was placed. |
| Sales.SalesOrderDetail | SalesOrderDetailID | INT | PK | NOT NULL, Identity | Line item identifier within a sales order. |
| Sales.SalesOrderDetail | SalesOrderID | INT | FK | | The sales order to which this detail belongs. |
| Sales.SalesOrderDetail | ProductID | INT | FK | | ID of the product being sold. |
| Sales.SalesOrderDetail | UnitPrice | MONEY | | NOT NULL | Price per unit of product. |
| Production.Product | ProductID | INT | PK | NOT NULL, Identity | Identifier for each product. |

| Table Name | Column Name | Data Type | PK/FK | Constraints | Description |
|---|---|---|---|---|---|
| Production.Product | Name | NVARCHAR(50) | | NOT NULL | Name of the product. |
| Production.Product | ProductNumber | NVARCHAR(25) | | NOT NULL | Unique product code or SKU. |

## Object Descriptions

| Object Type | Name | Description |
|---|---|---|
| Table | Person.Person | Stores personal details like name, contact info, and title. |
| Table | Sales.SalesOrderHeader | Represents customer sales orders. |
| Table | Production.Product | Contains data about items sold by the company. |
| View | Top10BestSoldProducts | Displays top 10 best-selling products based on total quantity sold, and how many distinct orders each appeared in. |
| Trigger | | Automatically updates the ModifiedDate when a record is changed. |
| Procedure | | Returns a list of orders for a specified customer ID. |

# Proposed Database Environment

## Database Platform

**SQL Server Management Studio 2022**

Our reasoning for choosing this platform is that SQL Server Management Studio 2022 supports enterprise-grade features like stored procedures, triggers, role-based security, and performance optimisation tools.

## Database Structure Overview

**Database Name:** AdventureWorks2022

Includes tables for:

- Person, Employee, Customer
- Product, ProductCategory, Supplier
- SalesOrder, SalesOrderDetail, PurchaseOrder, PurchaseOrderDetail
- Department, UserLogins, and logs such as ProductLog

Also Includes:

- Views for reporting
- Stored Procedures for dynamic access
- Triggers for change tracking
- Functions for business logic
- Indexes for performance

## Data File Organisation

### Primary Filegroup (PRIMARY)

- Contains core schema and primary data tables.

### Secondary Filegroups (optional, for performance)

- INDEXES: Store non-clustered indexes for large transactional tables (like SalesOrderDetail, PurchaseOrderDetail)
- ARCHIVE: Older order and log data can be moved here for storage efficiency.

### File Types:

- .MDF – Primary data file.
- .NDF – Optional secondary files (indexes, archived data).
- .LDF – Transaction log file for recovery and rollback.

## Security Requirements

### Authentication and Authorisation:

Mixed-mode authentication: (SQL & Windows).

### Role-Based Access Control (RBAC)

- SalesRole: SELECT/INSERT on sales tables/views
- HRRole: SELECT/UPDATE on employee records
- AdminRole: Full DB access
- AuditorRole: Read-only access for reports

### User Permissions:

- Use CREATE LOGIN and CREATE USER to grant least-privilege access.
- Sensitive data (e.g., salaries, contact info) restricted by views with filtering or column masking.

### Audit & Monitoring:

- ProductLog trigger captures stock changes.
- Use SQL Server Management Studio Audit (or PostgreSQL logging) to track access/modifications.
- Scheduled reports on changes to critical tables.

## Performance Requirements

### Indexing:

- Clustered indexes on primary keys (ProductID, SalesOrderID, etc.)
- Non-clustered indexes on frequently filtered fields (OrderDate, CustomerID, EmployeeID)
- Use covered indexes for reporting queries

### Query Optimisation:

- Use parameterised stored procedures instead of ad-hoc queries.
- Regular use of execution plan analysis for heavy queries.
- Avoid table scans by ensuring correct indexing.

### Data Volume Handling:

- Partition large tables by OrderDate or DepartmentID (optional).
- Archive old orders/purchases periodically to reduce load.

### Maintenance:

- Set up daily backups (full + differential + transaction log).
- Regular index maintenance: REBUILD or REORGANISE.
- DBCC CHECKDB for integrity checks.

## Physical Database Implementation

The Physical Database has been sent on Teams, and the query pages are in the Project folder.